

Heavy metal content of ayurvedic herbal medicine products.

CONTEXT: Lead, mercury, and arsenic intoxication have been associated with the use of Ayurvedic herbal medicine product (HMPs). OBJECTIVES: To determine the prevalence and concentration of heavy metals in Ayurvedic HMPs manufactured in South Asia and sold in Boston-area stores and to compare estimated daily metal ingestion with regulatory standards. DESIGN AND SETTING: Systematic search strategy to identify all stores 20 miles or less from Boston City Hall that sold Ayurvedic HMPs from South Asia by searching online Yellow Pages using the categories markets, supermarkets, and convenience stores, and business names containing the word India, Indian cities, and Indian words. An online national directory of Indian grocery stores, a South Asian community business directory, and a newspaper were also searched. We visited each store and purchased all unique Ayurvedic HMPs between April 25 and October 24, 2003. MAIN OUTCOME MEASURES: Concentrations (microg/g) of lead, mercury, and arsenic in each HMP as measured by x-ray fluorescence spectroscopy. Estimates of daily metal ingestion for adults and children estimated using manufacturers' dosage recommendations with comparisons to US Pharmacopeia and US Environmental Protection Agency regulatory standards. RESULTS: A total of 14 (20%) of 70 HMPs (95% confidence interval, 11%-31%) contained heavy metals: lead (n = 13; median concentration, 40 microg/g; range, 5-37,000), mercury (n = 6; median concentration, 20,225 microg/g; range, 28-104,000), and/or arsenic (n = 6; median concentration, 430 microg/g; range, 37-8130). If taken as recommended by the manufacturers, each of these 14 could result in heavy metal intakes above published regulatory standards. CONCLUSIONS: One of 5 Ayurvedic HMPs produced in South Asia and available in Boston South Asian grocery stores contains potentially harmful levels of lead, mercury, and/or arsenic. Users of Ayurvedic medicine may be at risk for heavy metal toxicity, and testing of Ayurvedic HMPs for toxic heavy metals should be mandatory.